

NURBEK TASTAN

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PROFILE

Doctoral researcher in machine learning with a strong foundation in probabilistic modeling, optimization, and scalable systems. My research focuses on **trustworthy federated learning**, with particular emphasis on **fairness**, **privacy**, **efficiency**, and **robustness** in collaborative settings. I also develop methods to make large-scale models, including **language models**, more efficiently fine-tunable while also preserving utility and user confidentiality.

Passionate about building machine learning systems that are not only performant, but also **equitable**, **privacy-aware**, and **computationally viable** for real-world deployment.

EDUCATION

PhD in Machine Learning, Mohamed bin Zayed University of Artificial Intelligence 2023 - Present
Abu-Dhabi, United Arab Emirates

Relevant Coursework: Foundations and Advanced Topics in Machine Learning, Advanced Probabilistic and Statistical Inference, Advanced Topics in Continuous Optimization, Federated Learning, Safe and Robust Computer Vision, Advanced Machine Learning

MSc in Machine Learning, Mohamed bin Zayed University of Artificial Intelligence 2021 - 2023
Abu-Dhabi, United Arab Emirates

Relevant Coursework: Mathematical Foundations for AI, Machine Learning, Advanced ML, Trustworthy AI, Probabilistic and Statistical Inference, Causality, Reinforcement Learning, Optimization

BSc in Systems of Information Security, International Information Technology University 2017 - 2021
Almaty, Kazakhstan

Relevant Coursework: Programming (Python, C++, Java), ML, Cybersecurity, a lot of Maths
GPA **3.94**/4.00 (Top 1st of 2021 graduating class (out of 900 students))

EXPERIENCE

Lead Teaching Assistant Jan 2022 - Present
MBZUAI *Abu-Dhabi, United Arab Emirates*

- Courses: [DS702: Big Data Processing], [MTH701: Mathematical Foundations of Artificial Intelligence]
- Led sessions for graduate courses, covering distributed data systems and mathematical foundations for AI.
- Designed problem sets and lab materials in probability theory, statistical inference, and scalable ML systems.
- Supervised student projects and delivered guidance on Spark, Hadoop, PageRank, LSH, and big data workflows.
- Enhanced student understanding through hands-on tutorials and assessment-driven feedback.
- More information on [this website](#).

Data Scientist May 2020 - Sep 2021
InCyberService (Healthcare) *Almaty, Kazakhstan*

- Developed classification models for highly imbalanced healthcare data.
- Built OCR-based object detection pipelines for extracting clinical information from scanned documents.
- Designed and maintained a big data pipeline using Apache Spark, Airflow, and Hadoop.

Python Developer and Teacher Jan 2018 - Mar 2019
Bolashak School, IITU *Almaty, Kazakhstan*

- Built a full-stack school management system using Python and Django to streamline internal workflows.
- Taught programming fundamentals and advanced topics (OOP, web scraping, algorithms) in Python and C++.

SKILLS

Languages	Python, C++, Scala, SQL (PostgreSQL, ClickHouse, Cassandra)
Frameworks	PyTorch, Tensorflow, Spark/PySpark, HuggingFace Transformers
Research Focus	Federated Learning, Decentralized Learning, Trustworthy Machine Learning, Efficient Fine-tuning of Large Models, Optimization, Distributed Optimization
Additional Tools	Big Data: Apache Spark, PySpark, Hadoop, Kafka, Airflow, Spark Streaming, Tools & Infrastructure: Docker, Git, Unix/Linux, REST APIs, TensorBoard, W&B

PUBLICATIONS

Aequa: Provably Fair Federated Learning using Slimmable Networks.

Nurbek Tastan, Samuel Horvath, Karthik Nandakumar. ICML, 2025. Under Review.

A Framework for Double-Blind Federated Adaptation of Foundation Models.

Nurbek Tastan, Karthik Nandakumar. ICLR 2025 Workshop on Modular, Collaborative, and Decentralized Deep Learning (MCDL@ICLR25). ICCV, 2025. Under Review.

CYCLE: Choosing Your Collaborators Wisely to Enhance Collaborative Fairness in Decentralized Learning.

Nurbek Tastan, Samuel Horvath, Karthik Nandakumar. TMLR. Under Review.

FedPeWS: Personalized Warmup via Subnetworks for Enhanced Heterogeneous Federated Learning.

Nurbek Tastan, Samuel Horvath, Martin Takac, Karthik Nandakumar. CPAL, 2025.

Redefining Contributions: Shapley-Driven Federated Learning.

Nurbek Tastan, Samar Fares, Toluwani Aremu, Samuel Horvath, Karthik Nandakumar. IJCAI, 2024.

Collaborative Learning of Anomalies with Privacy (CLAP) for Unsupervised Video Anomaly Detection: A New Baseline.

Anas Al-lahham, Muhammad Zaigham Zaheer, Nurbek Tastan, Karthik Nandakumar. CVPR, 2024.

A Coarse-to-Fine Pseudo-Labeling (C2FPL) Framework for Unsupervised Video Anomaly Detection.

Anas Al-lahham, Nurbek Tastan, Muhammad Zaigham Zaheer, Karthik Nandakumar. WACV, 2024.

CaPriDe Learning: Confidential and Private Decentralized Learning based on Encryption-friendly Distillation Loss.

Nurbek Tastan, Karthik Nandakumar. CVPR, 2023.

Valid and Invalid Bitcoin Transactions.

Saule Amanzholova, Nurbek Tastan, Kamila Kalkamanova, Amina Yessenalina. ACM ICEMIS, 2020.

Burglary Detection Framework for House Crime Control.

Nurbek Tastan, Abdul Razaque, Mohamed Frej, Saule Amanzholova, R. Ganda, F. Amsaad. IEEE ICCSA, 2019.

PROJECTS

Collaborative Learning with Robustness to Poisoning Attacks. Implemented several poisoning attacks (label flip, random, noise, backdoor) and developed a defense mechanism using anomaly detection over inter-client information. Designed to eliminate malicious participants in decentralized setups. Implemented in PyTorch.

Registration Plate Number Detection. Built a real-time video pipeline for license plate recognition using YOLO and Tesseract OCR. Processed live video, extracted frames, detected plates, and applied OCR to retrieve alphanumeric content. Deployed the system for access control automation.

Causality Problem: Cyber-Security Attacks on Graph Data. Developed a graph-based application to model and analyze cybersecurity attacks from vulnerability data (e.g., buffer overflow, gateway breaches). Applied Greedy Equivalence Search (GES) and Peter-Clark (PC) algorithms to infer causal DAGs for threat propagation analysis.

Density Estimation in Continuous Bayesian Networks. Explored uncertainty modeling through continuous Bayesian networks. Built models to estimate parametric PDFs using linear Gaussian assumptions and showed improved accuracy over standard methods via TVD-based evaluations.

CONFERENCE / WORKSHOP PRESENTATIONS

- CaPriDe Learning: Confidential and Private Decentralized Learning based on Encryption-friendly Distillation Loss. **CVPR**, Vancouver, Canada, Jun 2023.
- A Coarse-to-Fine Pseudo-Labeling (C2FPL) Framework for Unsupervised Video Anomaly Detection. **2nd MBZUAI CL Workshop**, Abu Dhabi, UAE, Dec 2023.
- Collaborative Learning of Anomalies with Privacy (CLAP) for Unsupervised Video Anomaly Detection: A New Baseline. **WACV**, Waikoloa, Hawaii, USA, Jan 2024.
- Redefining Contributions: Shapley-Driven Federated Learning. **CVPR**, Seattle WA, USA, Jun 2024.
- Confidential, Private, and Fair Decentralized Learning. **IJCAI**, Jeju, South Korea, Aug 2024.
- FedPeWS: Personalized Warmup via Subnetworks for Enhanced Heterogeneous Federated Learning. **ADIA Symposium**, Abu Dhabi, UAE, Nov 2024.
- A Framework for Double-Blind Federated Adaptation of Foundation Models. **CPAL**, Stanford, USA, Mar 2025.
- ICLR MCDL Workshop, Singapore, Apr 2025.

ACADEMIC SERVICES

PC / Reviewer (Conference): AAAI 2024, ICLR 2025, CVPR 2025, ICML 2025, ICCV 2025, NeurIPS 2025
Reviewer (Journal): IEEE Transactions on Mobile Computing (TMC)

CERTIFICATES

Machine Learning Specialization by DeepLearning.AI Coursera, Dec, 2022
– Supervised Machine Learning: Regression and Classification
– Advanced Learning Algorithms
– Unsupervised Learning, Recommenders, Reinforcement Learning
Cybersecurity by IBM Coursera, Jun, 2020

HONORS & AWARDS

National & Prestigious Awards

Presidential Scholarship, Ministry of Education, Kazakhstan 2020
Award “Altyn Belgi” (Golden Badge), Ministry of Education, Kazakhstan 2017
National Presidential Olympiad, Ministry of Education, Kazakhstan (I place - regional, III place - final) 2016
Republican Scientific Competition (Applied Mathematics) - III place (Bronze) 2016
National Scientific Competition (Computer Science) - II place (Silver) 2020, 2021
National Mathematics competition among students, Kazakhstan, IV place 2019

International & Technical Competitions

ICPC regional, II place 2019
“Hack The Space” Hackathon, Dubai, United Arab Emirates, “The Best Math Team” 2022

University- or Region-level Competitions

Scientific competition (Computer Science), IITU, Kazakhstan, II place 2021
Scientific competition (Computer Science), IITU, Kazakhstan, III place 2020
Mathematics competition, IITU, I place 2019
Mathematics competition, IITU, II place 2018
Physics Olympiad, II place (Silver) 2015
Regional scientific competition (Applied Mathematics), I place 2015
Mathematics Olympiad, I place (Gold) 2013

REFERENCES

Dr. Karthik Nandakumar	Dr. Samuel Horváth
Primary Supervisor	Secondary Supervisor
Associate Professor at MBZUAI & MSU	Assistant Professor at MBZUAI
✉ karthik.nandakumar@mbzuai.ac.ae, nandakum@msu.edu	✉ samuel.horvath@mbzuai.ac.ae